Data Handling During a Pandemic;

The Informatics of Collating COVID Testing Data from a Personal to National Scale

Sydney Lannin, Avery Barnett, Madeleine Kiel, Phebe Foley

MIRO Board

Relevant Texts Brought to the Conversation

Sydney

It took a pandemic, but the US finally has (some) centralized medical data
- This source describes the processes of a database through which covid testing data was collected and transmitted

Missing science: A scoping study of COVID-19 epidemiological data in the United States
- This discusses another database that was built to collect and transmit testing data

A Contact Tracing-Health Information Exchange Partnership: An Effective Collaboration for a Data-Driven Response to COVID-19 in Maryland
- This source takes a closer at how Mayland handled the influx of covid testing information

Phebe

Challenges in reported COVID-19 data: best practices and recommendations for future epidemics
- This research paper identifies the best and worst practices for nationwide informatics during COVID, and how we could improve for future epidemics.

COVID-19 Case Surveillance: Trends in Person-Level Case Data Completeness
- This report highlights the level of data completeness of in person national COVID data and where inconsistencies and gaps in data stem from.

Information technology in emergency management of COVID-19 outbreak
- This research paper discusses the tools and technologies used in the collection and distribution of COVID data on the regional and national levels.

Avery

Remote Data Collection for Public Health Research
- This paper discusses the switch over to remote data collection of COVID testing, which was very present at the time in public health research.

John Hopkins: Data Collection Process, Challenges faced, and Lessons learned
- This journal shows the technical and logistical side of how the John Hopkins University COVID dashboard standardized data in real-time monitoring.

Standardizing, Harmonizing, and Protecting Data Collection…
- These researchers explain how standardized common data elements (CDE's) can help with data collection of COVID testing on a national scale, improving public health responses.

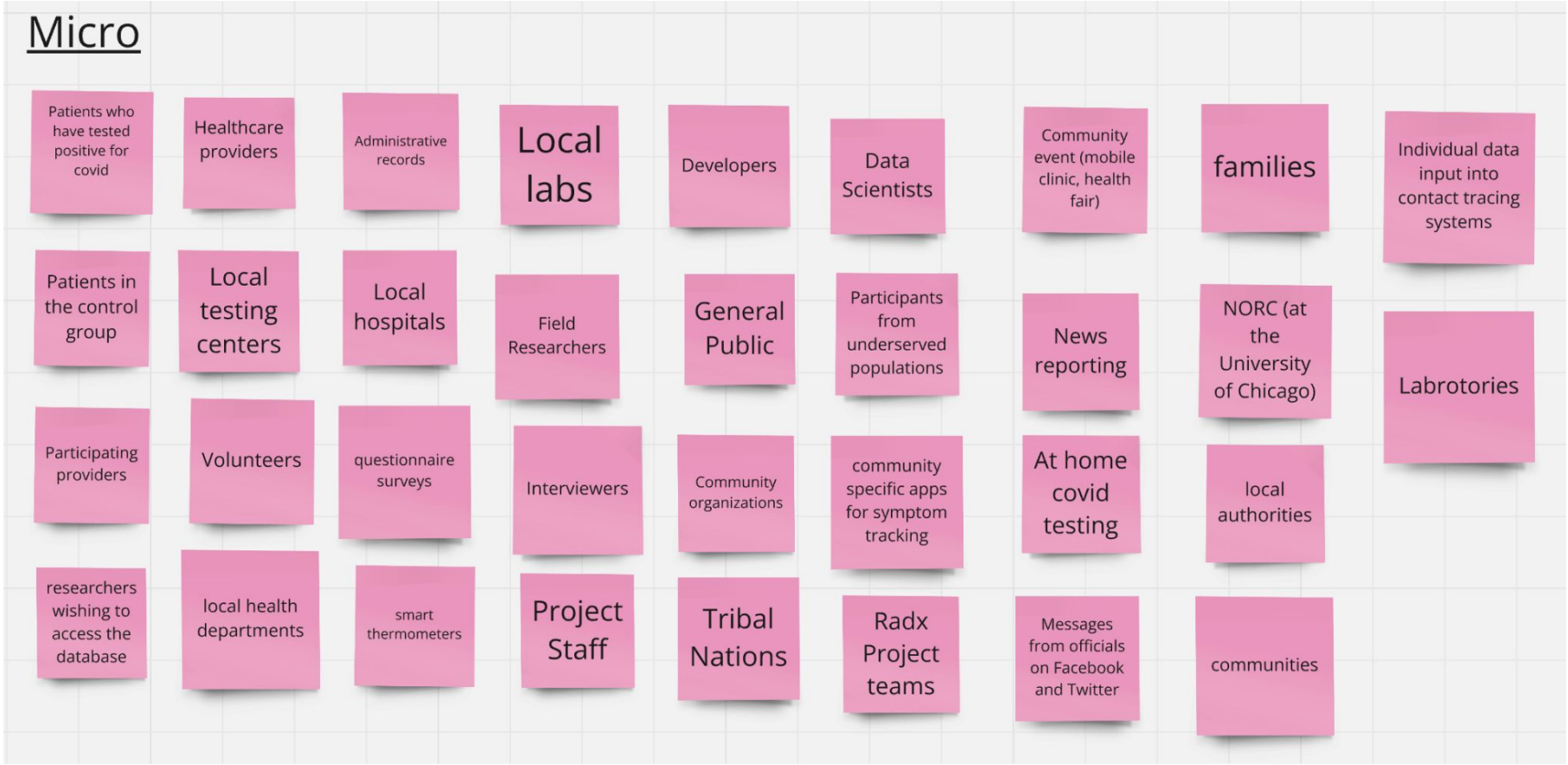
Madeleine

Distribution of COVID-19 Home Testing Through Community Health Centers: Results of the COVID CoNOce MÁS Study
- This research paper assess the acceptability and practicality of at home COVID-19 testing and reporting test results in community health centers

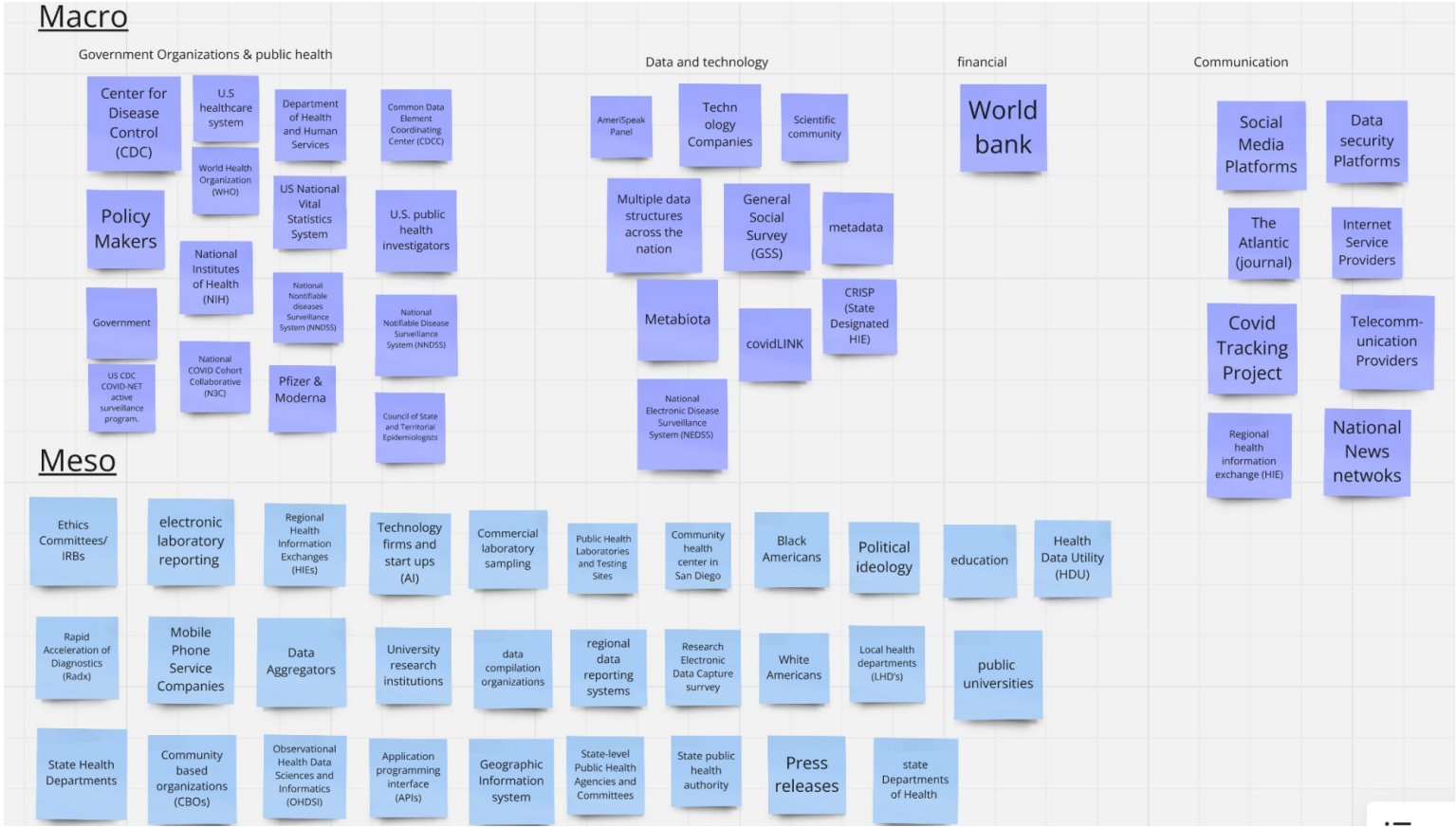
Estimating population infection rates from non-random testing data: Evidence from the COVID-19 pandemic
- Estimate population infection rates using non-random testing data to see population infection rates and analyze the relationship between and size of tested population and understand selection bias in results

COVID-19 pandemic's impact on the public's confidence in scientific community: The role of personal contact
- This Article addresses the public's confidence in the the scientific communities ability to report data concerning COVID-19 and its test results

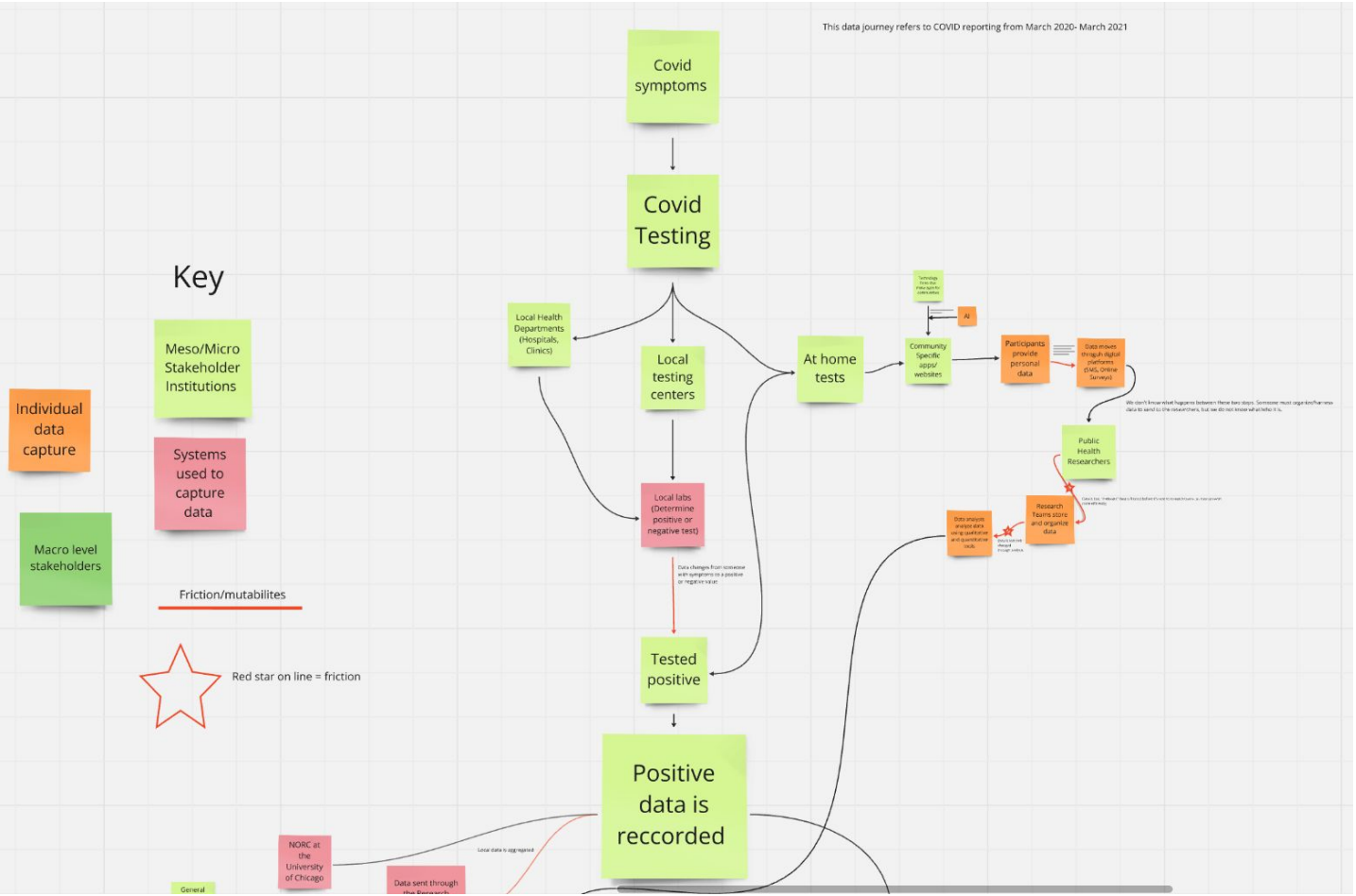
Merged Social Worlds Ecosystem (Micro)

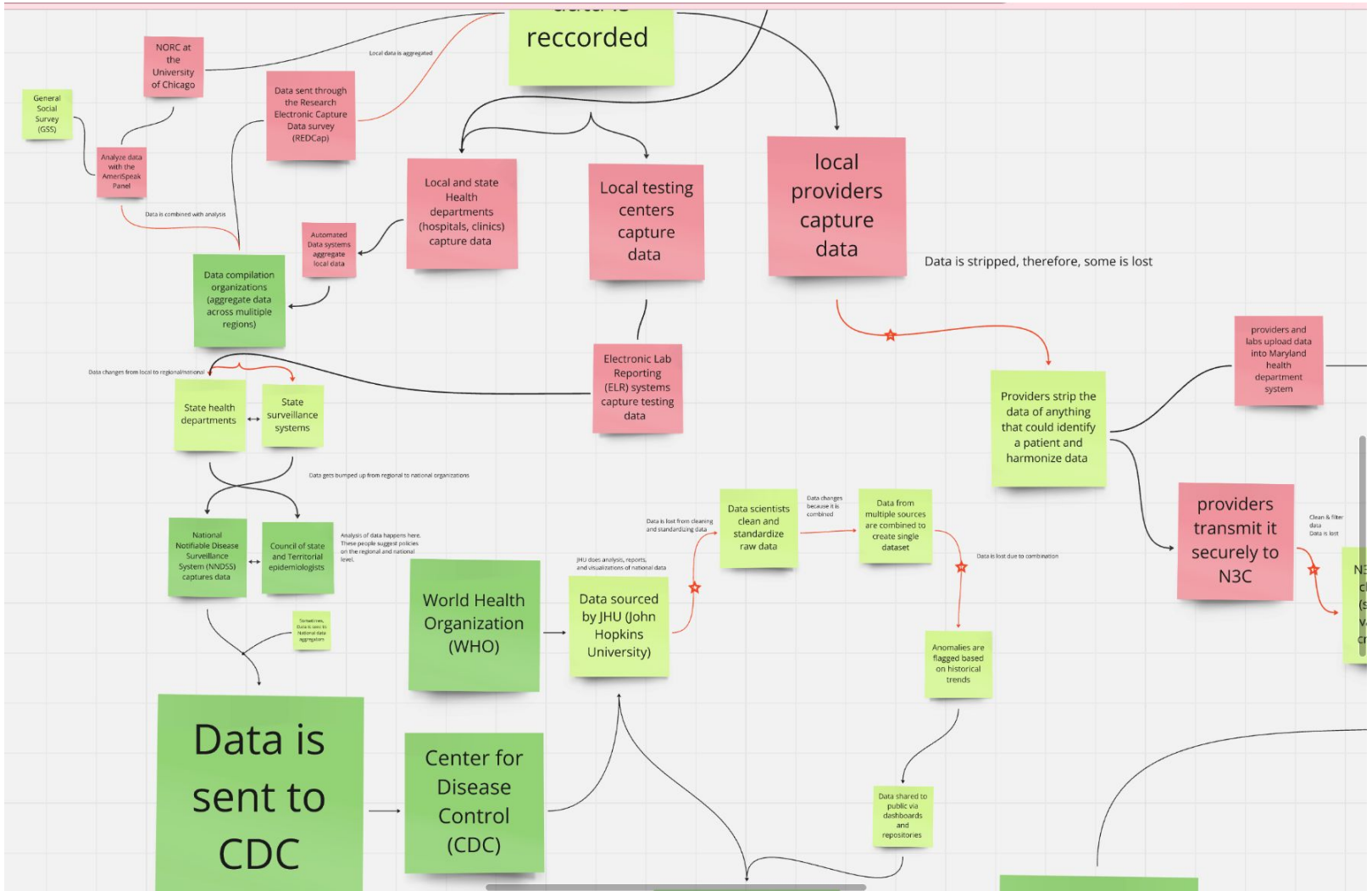


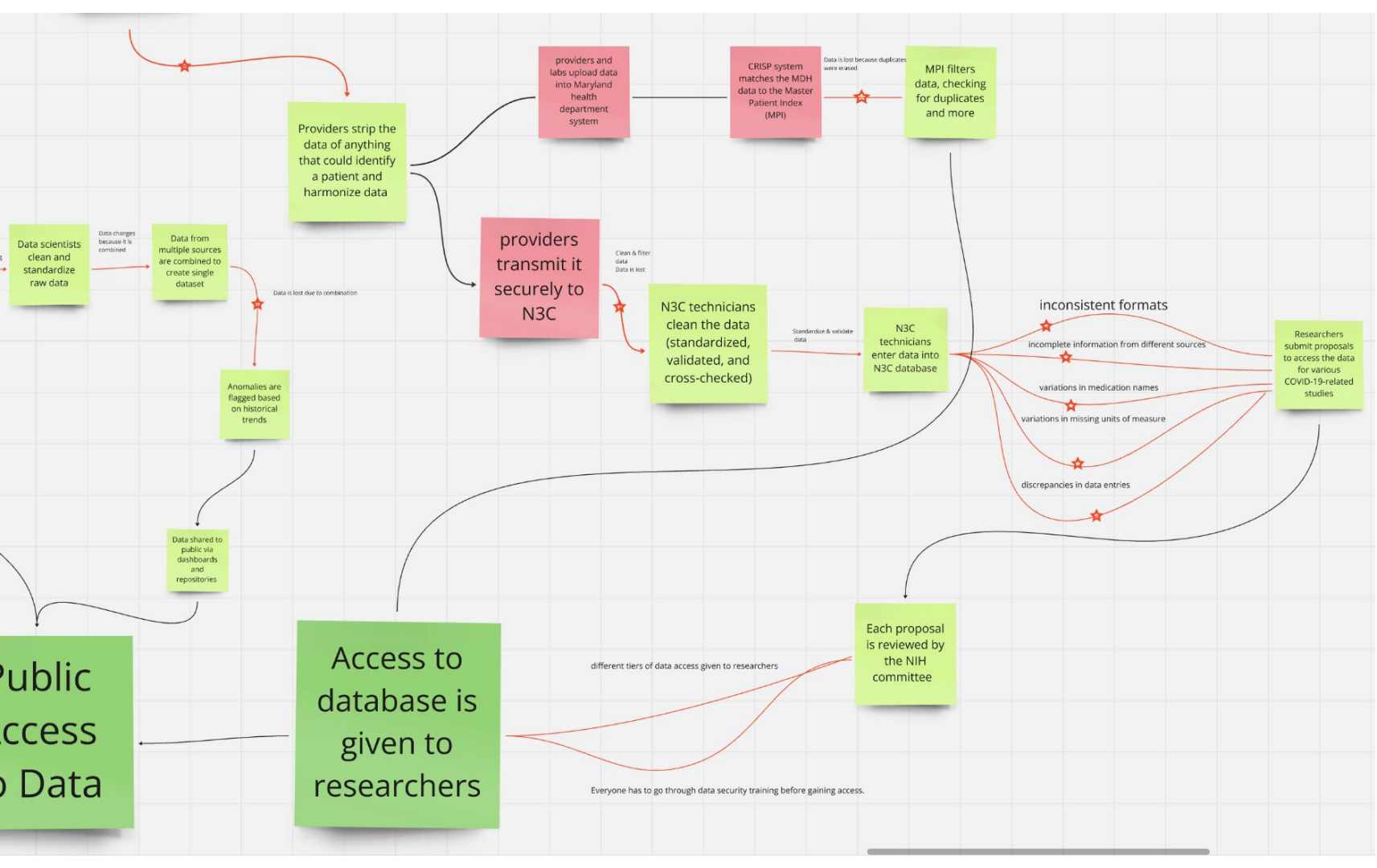
Merged Social Worlds Ecosystem (Macro and Meso)

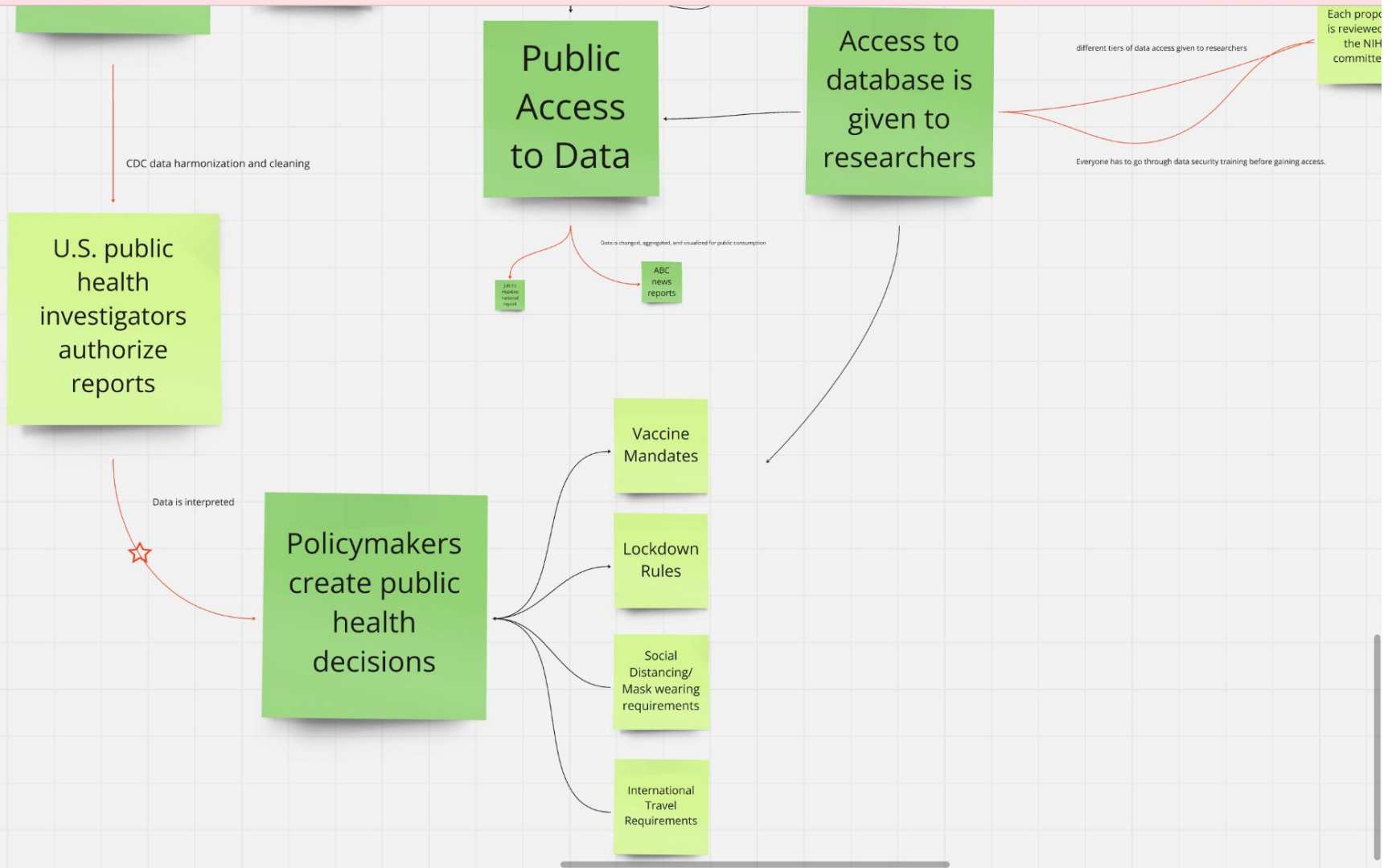


Merged Data Journey




reccorded
NORC at the University of Chicago
Local data is aggregated
General Social Survey (GSS)
Data sent through the Research Electronic Capture Data survey (REDCap)
Analyze data with the AmeriSpeak Panel
Local and state Health departments (hospitals, clinics) capture data
Local testing centers capture data
local providers capture data
Data is stripped, therefore, some is lost
Data is combined with analysis
Automated Data systems aggregate local data
Data compilation organizations (aggregate data across multiple regions)
Data changes from local to regional/national
Electronic Lab Reporting (ELR) systems capture testing data
providers and labs upload data into Maryland health department system
State health departments
State surveillance systems
Providers strip the data of anything that could identify a patient and harmonize data
Data gets bumped up from regional to national organizations
National Notifiable Disease Surveillance System (NNDSS) captures data
Council of state and Territorial epidemiologists
Analysis of data happens here. These people suggest policies on the regional and national level.
Data is lost from cleaning and standardizing data
Data scientists clean and standardize raw data
Data changes because it is combined
Data from multiple sources are combined to create single dataset
providers transmit it securely to N3C
Clean & filter data
Data is lost
JHU does analysis, reports, and visualizations of national data
World Health Organization (WHO)
Data sourced by JHU (John Hopkins University)
Data is lost due to combination
Anomalies are flagged based on historical trends
Sometimes, Data is sent to National data aggregators
Data is sent to CDC
Center for Disease Control (CDC)
Data shared to public via dashboards and repositories


providers and labs upload data into Maryland health department system
CRISP system matches the MDH data to the Master Patient Index (MPI)
Data is lost because duplicates were erased
MPI filters data, checking for duplicates and more
Providers strip the data of anything that could identify a patient and harmonize data
Data scientists clean and standardize raw data
Data changes because it is combined
Data from multiple sources are combined to create single dataset
Data is lost due to combination
Anomalies are flagged based on historical trends
Data shared to public via dashboards and repositories
providers transmit it securely to N3C
Clean & filter data
Data is lost
N3C technicians clean the data (standardized, validated, and cross-checked)
Standardize & validate data
N3C technicians enter data into N3C database
inconsistent formats
incomplete information from different sources
variations in medication names
variations in missing units of measure
discrepancies in data entries
Researchers submit proposals to access the data for various COVID-19-related studies
Each proposal is reviewed by the NIH committee
Access to database is given to researchers
different tiers of data access given to researchers
Everyone has to go through data security training before gaining access.
Public Access to Data


Public Access to Data
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CDC data harmonization and cleaning
U.S. public health investigators authorize reports
Data is changed, aggregated, and visualized for public consumption
ABC news reports
Data is interpreted
Policymakers create public health decisions
Vaccine Mandates
Lockdown Rules
Social Distancing/ Mask wearing requirements
International Travel Requirements

Notable Friction and Mutabilities

- Technological Disparities
  - The United States did not have a centralized way of collecting the massive amounts of testing data. This resulted in varied technologies and databases used to capture testing data, and because of this data formats were inconsistent, there were severe delays in transmitting this data, it was difficult to combine data, and more.
- Data Standardization and Cleansing
  - The processing of testing data included stripping and cleaning the data; anonymizing, validating, formatting for consistency, checking for duplicates, combining the data, and more. These processes, however, resulted in massive loss of data, and took a significant amount of time to do.
- Privacy Concerns
  - Covid patients may not feel comfortable with their data being shared with the public. This may create an obstacle as individuals try to protect their personal information.

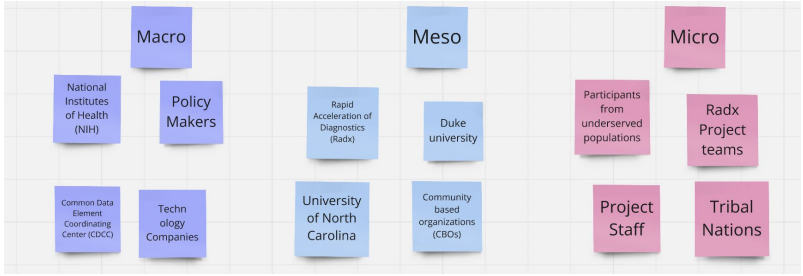
Appendix- Social World Diagrams (Avery Barnett)



[Remote data collection for public health research in a COVID-19 era: ethical implications, challenges and opportunities]



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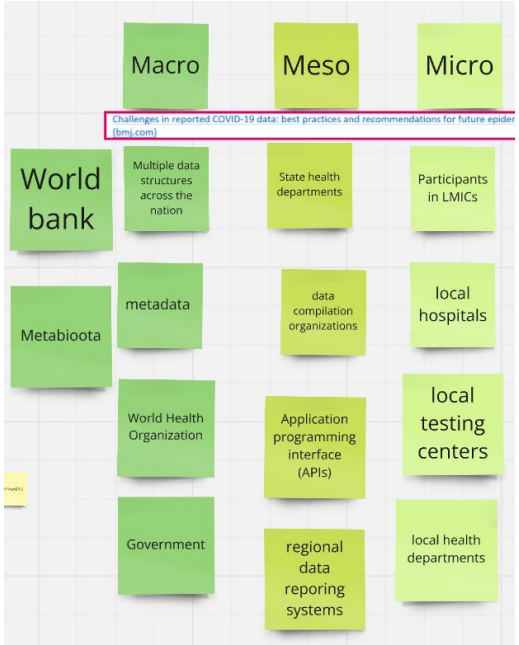


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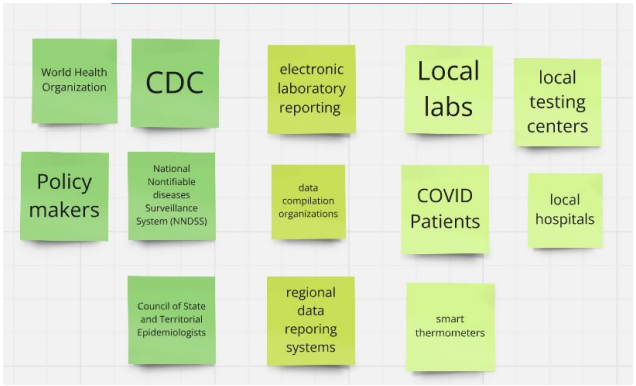
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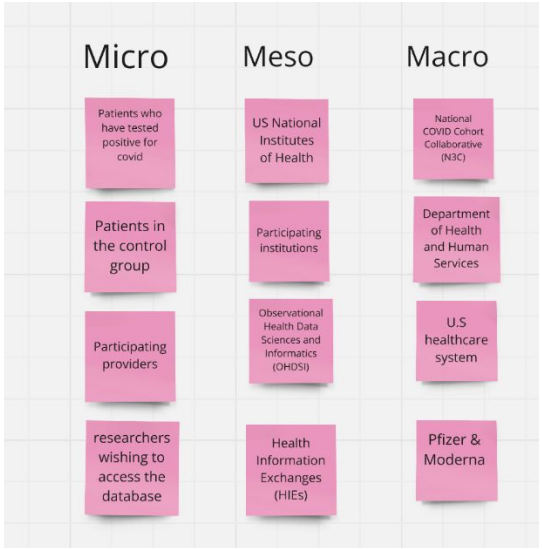


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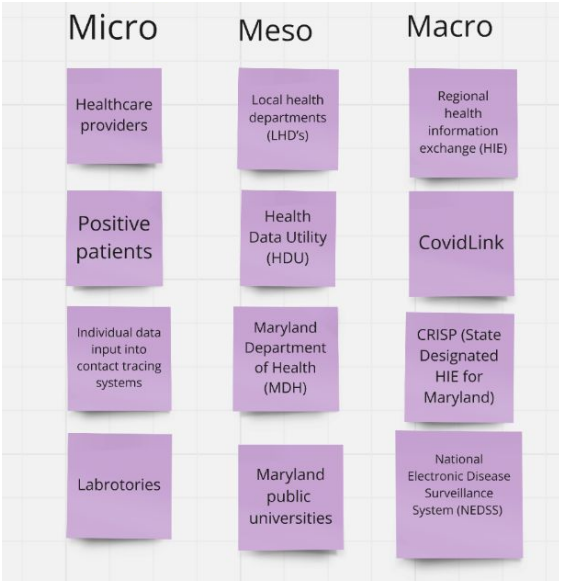
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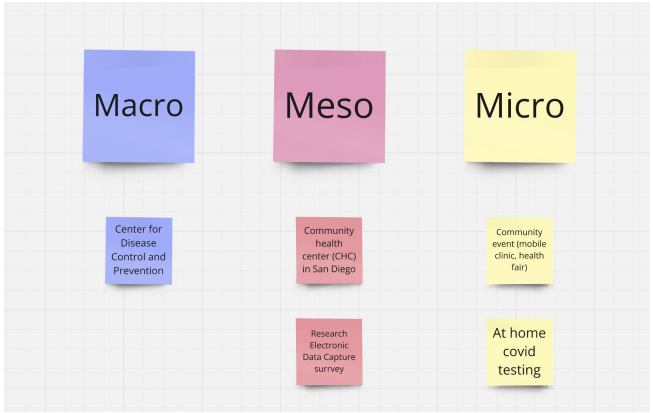


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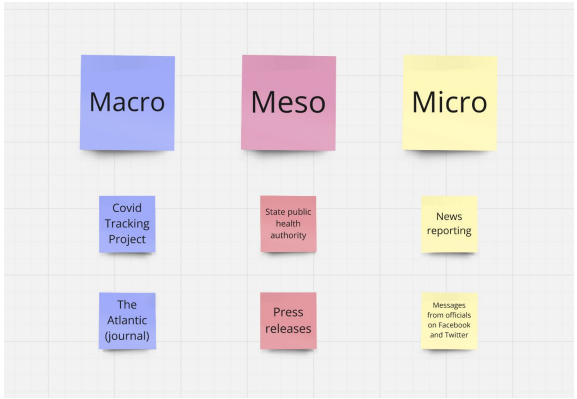


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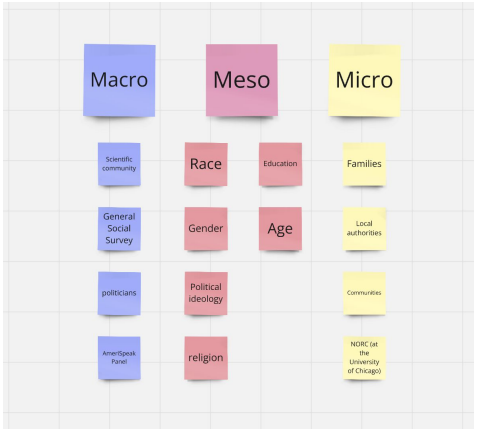
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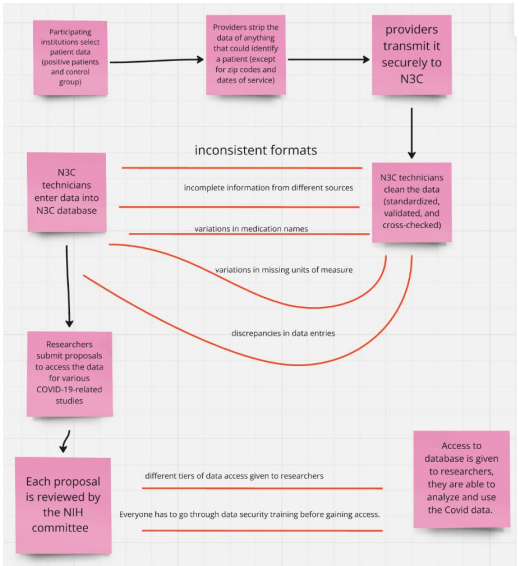


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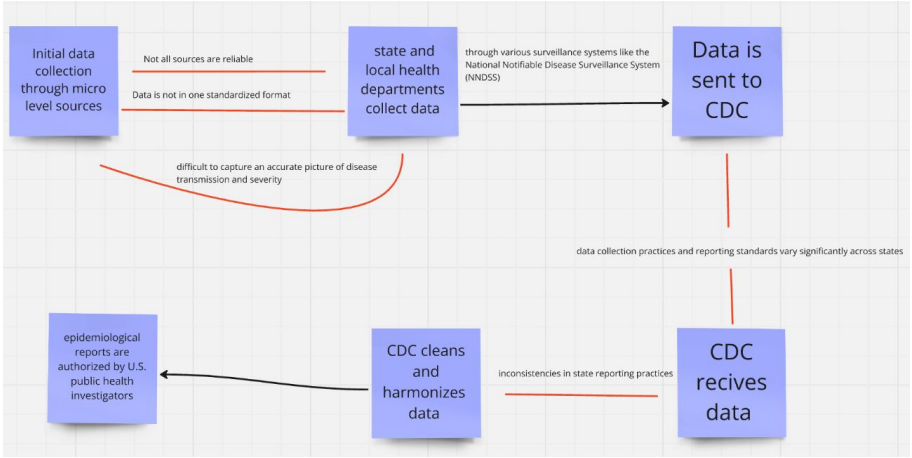


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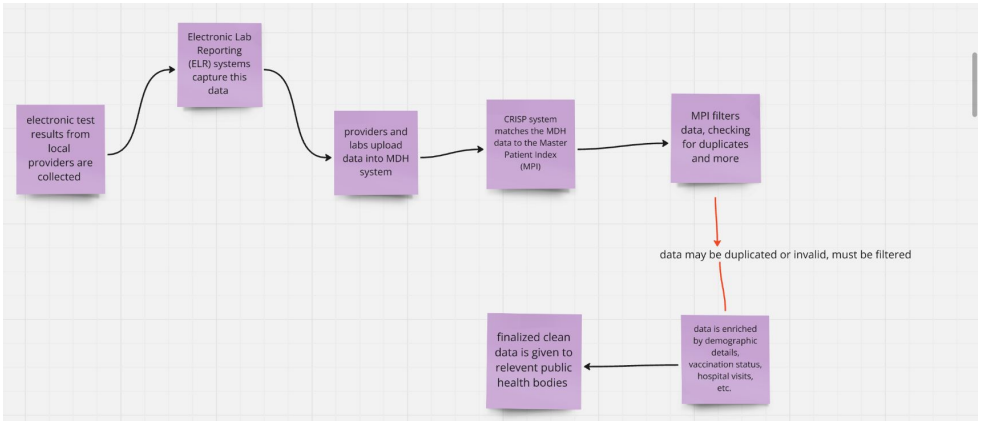
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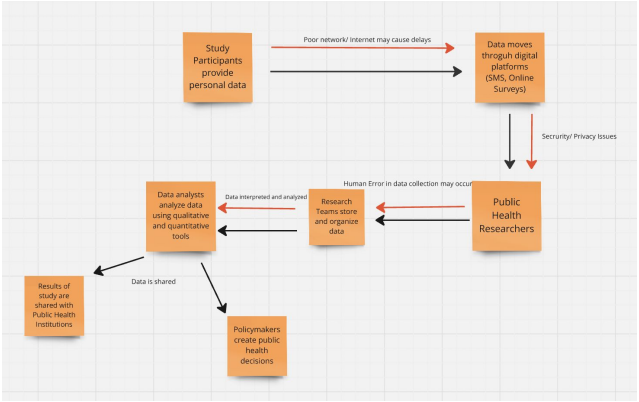
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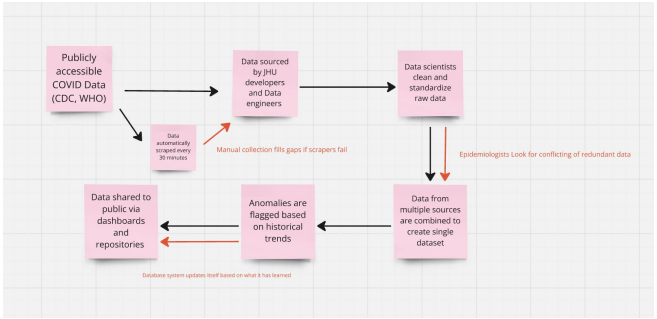
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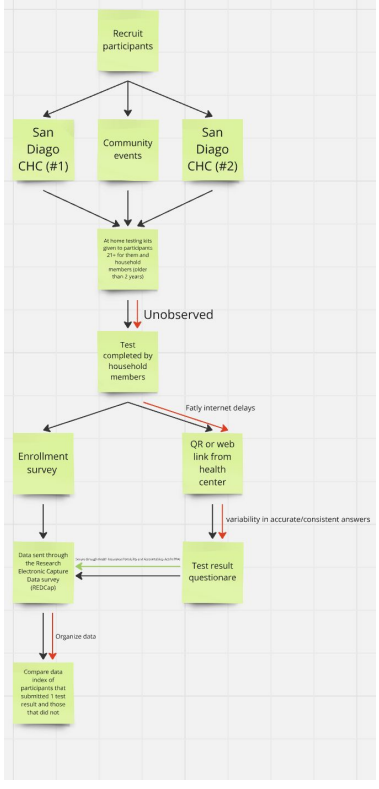


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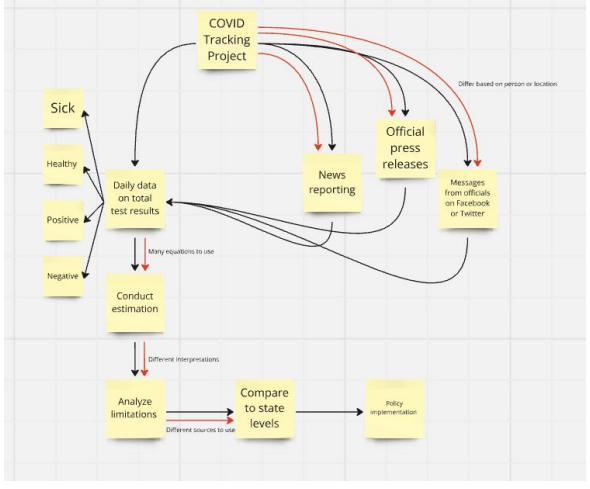


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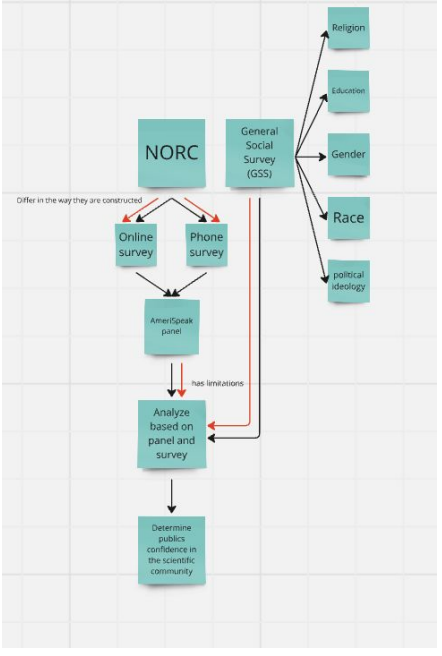
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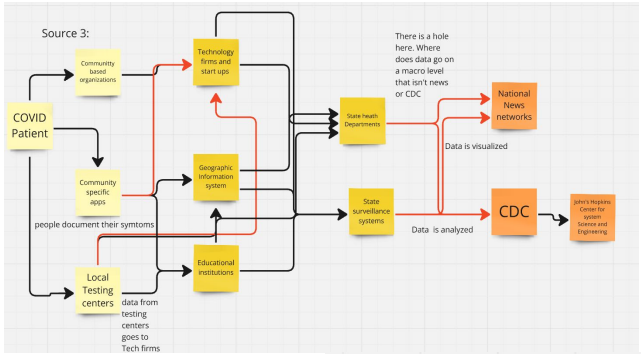


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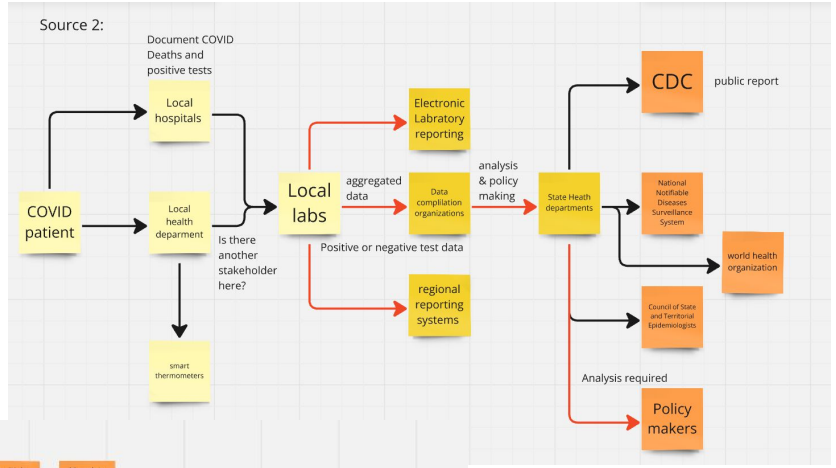
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